# Case Study Questions (Answer all)

## 1. Case Study – Banking Transactions (Batch Sequential)

A bank processes all daily transactions (withdrawals, deposits, transfers) at night using a batch job.
The system performs the following steps:

1. Validate all transaction records
2. Sort validated records by account number
3. Update the master account file
4. Generate summary reports

a. Identify each subsystem in this batch sequential process.
b. Explain why the Batch Sequential Architecture is suitable for this scenario.
c. Draw a simple block diagram showing the data flow.
d. State one limitation of using batch processing here.

## 2. Case Study – Real-Time Audio Processing (Pipe & Filter)

An audio application processes live microphone input by passing it through:

1. Noise Reduction
2. Equalizer
3. Compressor

a. Identify the filters and pipes in this system.
b. Explain how concurrency can improve performance in this pipeline.
c. State one benefit and one limitation of Pipe & Filter architecture.
d. Suggest one situation where this architecture may not work well.

## 3. Case Study – Traffic Light Control System (Process Control)

A city uses an automated traffic light controller to manage traffic flow at a busy four-way intersection. The system uses vehicle sensors embedded in the road, pedestrian crossing buttons, and timers. The controller adjusts the duration of green, yellow, and red lights based on real-time traffic conditions.

a. Identify the **input variables**, **control variables**, and **manipulated variables** in this system.
b. Define the term **set point** in the context of this traffic light control system.
c. Explain how **feedback** is used in this closed-loop control system.
d. Describe one **real-world disturbance** that could affect the system's performance.